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មូលន័យសង្ខេប

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ABSTRACT

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LIST OF ABBREVIATIONS

Abbreviation	Explanation
AI	Artificial Intelligence

Update with new abbrs

I. INTRODUCTION

The internship program is a cornerstone of the bachelor's curriculum at the **Institute of Digital Technology (IDT), Cambodia Academy of Digital Technology (CADT)**. Designed to bridge the gap between academic learning and professional practice, the program places students within real organizations where they can contribute to and learn from active projects in their field of study. This report documents the work undertaken, the challenges encountered, and the knowledge gained throughout the entirety of my second internship period.

1.1. Presentation of the Internship

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1.1.1. Objective of Internship

The primary objective of this internship, as defined by the IDT program guidelines, is to provide students with a structured opportunity to apply the theoretical knowledge and technical skills developed during their academic studies to the demands of a real professional environment. The program is intended to expose students to industry workflows, workplace communication norms, and the practical constraints that shape software development in an institutional context.

Beyond technical growth, the internship serves as a platform for students to evaluate their own professional strengths and identify areas for continued development. For the host organization, it provides the opportunity to assess a student's technical competency, initiative, and collaborative ability, a mutual exchange that may lead to future employment or continued collaboration.

1.1.2. Duration of Internship

The internship commenced on January 5th, 2026, and concluded on July 5th, 2026, spanning a total of six months (24 weeks). In accordance with the CADT Year 4 curriculum requirements, a minimum of 960 hours of documented professional work was completed over this period.

1.2. Presentation of the Company

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1.2.1. General Information of the Company

The internship took place at the **Lorem ipsum dolor sit amet, consectetur.** Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magnam aliquam quaerat voluptatem. Ut enim aequale doleamus animo, cum corpore dolemus, fieri tamen permagna accessio potest, si aliquod aeternum et infinitum impendere malum nobis opinemur. Quod idem licet transferre in voluptatem, ut postea variari voluptas distinguere possit, augeri amplificarique non possit. At etiam Athenis, ut e patre audiebam facete et urbane Stoicos irridente, statua est in quo a nobis philosophia defensa et. [Figure 1](#) illustrates the company's logo.



Figure 1: Lorem ipsum dolor sit amet, consectetur.'s logo

1.2.2. Vision and Mission

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1.2.3. Organizational Chart

Figure 2 illustrates the institutional hierarchy of the Lorem ipsum dolor sit amet, consectetur. The organization is led by the President of the lorem, supported by the Vice-President.

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> look inside
> it's empty



Figure 2: Lorem ipsum dolor sit amet, consectetur: 's organization chart

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1.2.4. Address and Contact

The Lorem ipsum dolor sit amet, consectetur. is located at:

- Address: [Somewhere on earth i think](#)
- Phone: (+855) 12 345 678 / (+60) 32 303 198 1
- Website: www.goon.edu.my
- Email: info@goon.edu.my

II. PROJECT DEFINITION AND SCOPE

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2.1. Problematic

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2.2. Project Objective

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2.3. Targeted Impact

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2.4. Methodology

The **lorem** project followed an **adaptive, sprint-based development approach** inspired by the Scrum framework. Unlike a rigid waterfall process where requirements are fully specified before development begins.

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2.4.1. Sprint Structure

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2.4.2. Task and Backlog Management

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2.4.3. Roles and Collaboration

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III. LITERATURE REVIEW

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3.1. Existing Systems Analysis

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3.2. Engineering Theory and Design Patterns

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3.3. Summary

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IV. PROJECT ANALYSIS AND CONCEPTS

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4.1. Requirements Specification

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4.1.1. Functional Requirements

The functional scope is [Table 1](#) lists the functional requirements across these domains.

Table 1: Functional requirements of the lorem system

ID	Feature	Description
FR-01	Lorem ipsum dolor.	Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua quaerat voluptatem. Ut enim aequi.
FR-02	Lorem ipsum dolor.	Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua quaerat voluptatem. Ut enim aequi doleamus animo, cum corpore dolemus, fieri tamen permagna accessio potest.
FR-03	Lorem ipsum dolor.	Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua quaerat voluptatem. Ut enim aequi doleamus animo, cum corpore dolemus, fieri tamen permagna.

ID	Feature	Description
FR-04	Lorem ipsum dolor.	Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magnam aliquam quaerat voluptatem. Ut enim aequaeque doleamus animo, cum corpore dolemus, fieri tamen permagna accessio potest, si aliquod aeternum et infinitum impendere malum nobis opinemur. Quod idem licet.
FR-05	Lorem ipsum dolor.	Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magnam aliquam quaerat voluptatem. Ut enim aequaeque doleamus animo, cum corpore dolemus, fieri tamen.
FR-06	Lorem ipsum dolor.	Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut.
FR-07	Lorem ipsum dolor.	Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magnam aliquam quaerat voluptatem. Ut.

4.1.2. Non-Functional Requirements

Table 2 Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua quaerat voluptatem. Ut enim aequale doleamus animo, cum corpore dolemus, fieri tamen permagna accessio potest, si.

Table 2: Non-functional requirements

Feature	Description
Security	The backend should enforce JWT authentication, HTTPS, rate limiting, input validation, returning structured error codes that do not leak internal details. The client should keep tokens in platform secure storage.
Performance	The backend should enforce server-side pagination on all list endpoints to prevent unbounded data transfer and offload attachment delivery to the CDN so large transfers do not degrade API response times.
Reliability and Availability	The backend should maintain persistent IMAP IDLE connections with automatic reconnection. The client should handle network failures gracefully with clear feedback and retry options.
Maintainability	Both codebases should follow a modular architecture: NestJS modules, controllers, and guards on the backend, and a clear separation of data models, services, state management, and UI on the client.
Scalability	The design should support future additions such supporting additional providers, investor tagging, analytics, and advanced AI modules.
Usability and Localization	The mobile interface should be simple, readable, and optimized for non-technical and technical staff, with important actions reachable in minimal taps, and should support Khmer and English with room for more languages.
Testability and Delivery	The backend should include automated end-to-end and integration tests for critical flows (authentication, mail fetching, permission resolution, contact creation), with delivery managed through a CI pipeline and automated database migrations.

4.2. Use Case Analysis

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4.2.1. User Account

Figure 3 Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magnam aliquam quaerat voluptatem. Ut enim aequi doleamus animo, cum corpore dolemus, fieri tamen permagna accessio potest, si aliquod aeternum et infinitum impendere malum nobis opinemur. Quod idem licet transferre.

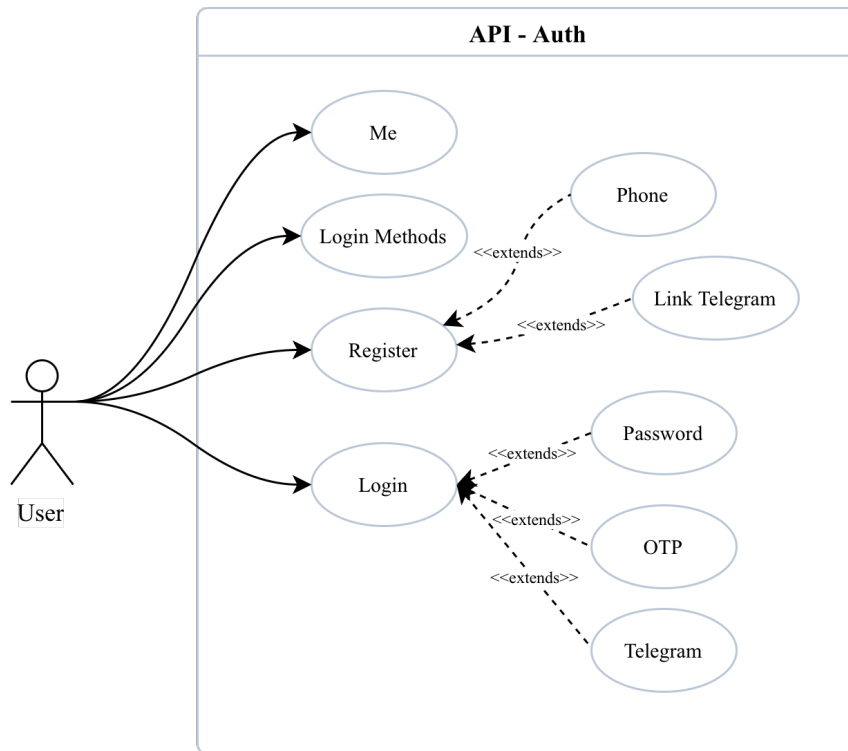


Figure 3: User account use case diagram

4.3. Process Flow and Activity Concepts

This section describes the main activity flow of the system from login to communication and follow-up, covering both client-side interactions and backend processing steps.

4.3.1. Core Workflow

1. Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magnam aliquam quaerat voluptatem. Ut enim aequi doleamus animo, cum corpore dolemus, fieri tamen.
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4.3.2. Activity Diagram



Activity Diagram for Main User Workflow

Figure 4: [TODO] Activity Diagram for Main User Workflow

4.4. Project Timeline



Table 3: Activities timeline

Tasks	Weeks																							
	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24
Requirement Analysis																								
Setup Project																								
Development Phase																								
Testing																								
Bug Fixes and UAT																								
Documentation																								

Table 3 outlines the delivery schedule across the 24-week internship period. Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua quaerat voluptatem. Ut enim aequale doleamus animo, cum corpore dolemus, fieri tamen permagna accessio potest, si aliquod aeternum et infinitum impendere malum nobis opinemur. Quod idem licet transferre in voluptatem, ut postea variari voluptas distinguere possit, augeri amplificarique non possit. At etiam Athenis, ut e patre audiebam facete et urbane Stoicos irridente, statua est in quo a nobis philosophia defensa et.

V. DETAIL CONCEPT

This chapter presents the detailed concept of the **lorem** system. While the previous chapter focused on analysis and requirements, this chapter translates those findings into a concrete design structure: the logical architecture, key design decisions, the data model, the data flow, the physical architecture, and the technologies required for implementation.

5.1. Logical Architecture

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5.1.1. System Overview

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Component diagram



5.2. Key Design Decisions

This section documents the decisions made before and during implementation that shaped the system's architecture. Each entry records the rationale behind the chosen approach; implementation details are covered in [Section VI](#).

5.3. Data Model Design

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Throughout this chapter and [Section VI](#), table names are written in `schema.table` notation; for example, `entity.person_entity` refers to the `person_entity` table in the PostgreSQL `entity` schema, reflecting the database's use of named schemas to group related tables by domain.

5.3.1. Entity Overview

The system's database organizes persistent data around five core entity groups. [Figure 5](#) shows their relationships.

"ERD: entity-relationship diagram",

Figure 5: [TODO] database entity-relationship diagram

5.4. Physical Architecture

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5.4.1. Deployment Overview

The physical architecture consists of the following main nodes:



Deployment diagram: physical nodes



5.5. Tools and Technology Requirements

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5.5.1. Common Technologies

a. Firebase Cloud Messaging (FCM)

Firebase Cloud Messaging is the push notification infrastructure used to alert officials of new incoming mail in real time.¹ Integrated into both the NestJS Worker process and the Flutter client, FCM handles the reliable delivery of background alerts and foreground data messages. The client uses FCM routing logic to intercept notification taps and navigate the user directly to the relevant email detail screen.

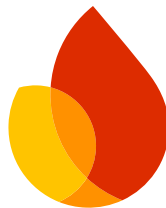


Figure 6: Firebase logo

b. JSON Web Tokens (JWT)

JSON Web Tokens are a compact, URL-safe standard for representing claims between two parties. A signed token carries a verifiable payload without requiring a server-side session lookup on every request, making it well suited for stateless API authentication. A dual-token strategy using short-lived access tokens and longer-lived refresh tokens is a common pattern for balancing security and user experience.

¹<https://firebase.google.com>

5.5.2. Backend Technologies

a. NestJS (TypeScript)

NestJS is a Node.js framework for building server-side applications in TypeScript.² It provides a modular architecture with native dependency injection, decorator-based guards and interceptors, and built-in support for REST APIs, WebSockets, and background workers, making it suitable for complex backend systems that require clear separation of concerns.



Figure 7: NestJS logo

b. PostgreSQL

PostgreSQL is an open-source relational database known for its extensibility and standards compliance. It supports advanced features including native full-text search via `TSVECTOR`, semi-structured data storage via `JSONB`, and strong ACID guarantees across complex relational schemas.

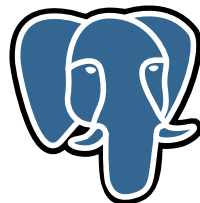


Figure 8: PostgreSQL logo

c. Sequelize

Sequelize is a promise-based ORM for Node.js that supports multiple relational databases including PostgreSQL.³ It provides model definitions, associations, query building, and a migration system through a TypeScript API, reducing the need to write raw SQL for most database operations.

²<https://nestjs.com>

³<https://sequelize.org>



Figure 9: Sequelize logo

d. Redis

Redis is an open-source in-memory data store used as a cache, database, and message broker.⁴ Its sub-millisecond read and write performance makes it well-suited for use cases requiring high throughput and low latency, such as session storage, rate limiting, and job queue backends.



Figure 10: Redis logo

e. BullMQ

BullMQ is a Node.js job queue library built on top of Redis.⁵ It provides named queues with support for concurrency control, retry logic, job prioritization, and lifecycle events, making it suitable for managing background task pipelines where reliability and ordering are important.



Figure 11: BullMQ logo

f. ImapFlow

ImapFlow is a Node.js IMAP client library that provides a modern Promise-based API for communicating with mail servers.⁶ It includes first-class support for the IMAP IDLE extensio, which enables real-time server-pushed mail event notifications without polling.

⁴<https://redis.io>

⁵<https://docs.bullmq.io>

⁶<https://imapflow.com>



Figure 12: ImapFlow logo

5.5.3. Frontend Technologies

a. Flutter (Dart)

Flutter is a cross-platform UI framework developed by Google that compiles a single Dart codebase to native iOS, Android, and web targets.⁷ Its hardware-accelerated rendering engine draws UI directly to a canvas rather than relying on native components, producing consistent visual output across all supported platforms.



Figure 13: Flutter logo

b. Telegram WebApp API

The Telegram WebApp API allows web applications embedded inside Telegram to access platform context such as the user's current theme parameters, viewport dimensions, and native UI behaviors including haptic feedback and the hardware back button.⁸



Figure 14: Telegram logo

c. GoRouter

GoRouter is a declarative routing package for Flutter built on top of the Navigator 2.0 API.⁹ It provides a URL-based route tree with support for nested routes, route guards via redirect callbacks, and deep link handling, making it suitable for applications with complex navigation requirements and authentication boundaries.

⁷<https://flutter.dev>

⁸<https://core.telegram.org/bots/webapps>

⁹https://pub.dev/packages/go_router

d. Provider

Provider is a state management library for Flutter built on top of the InheritedWidget mechanism.¹⁰ It allows application state to be exposed to the widget tree in a structured way, with scoped rebuilds limited to widgets that explicitly depend on the changed value.

e. Flutter Localization (l10n / ARB)

Flutter's built-in localization framework manages multilingual string resources via ARB files, with generated delegate classes providing type-safe access to translated strings throughout the widget tree.¹¹ The active locale is resolved at runtime and can be changed without restarting the application.

f. Freezed and json_serializable

Freezed is a code generation library for Dart that produces immutable value classes with built-in equality, copying, and pattern matching.¹² Paired with json_serializable, it generates type-safe JSON serialization and deserialization logic from annotated class definitions, eliminating manual mapping between API response shapes and application data models.

5.5.4. Development Tools

a. Visual Studio Code

Visual Studio Code is the primary development environment used for both the Flutter client and the NestJS backend. It provides Flutter and Dart plugin support including hot reload and widget inspection, as well as a strong TypeScript and NestJS plugin ecosystem for backend development.

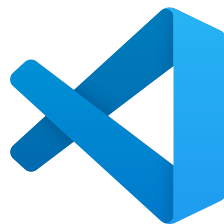


Figure 15: Visual Studio Code logo

¹⁰<https://pub.dev/packages/provider>

¹¹<https://docs.flutter.dev/ui/accessibility-and-internationalization/internationalization>

¹²<https://pub.dev/packages/freezed>

b. NestJS CLI

The NestJS CLI is a command-line tool for scaffolding and managing NestJS projects.¹³ It provides commands for generating modules, controllers, services, and guards following the NestJS convention, and manages the monorepo build configuration natively.

c. Git

Git is used for version control across both the Flutter and NestJS codebases. Branch-based development allows features and fixes to be isolated from the main branch until reviewed, and the commit history provides a traceable record of design decisions made throughout the internship period.

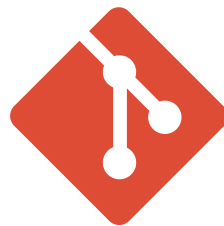


Figure 16: Git logo

d. GitLab

.....¹⁴



Figure 17: GitLab logo

e. Postman

Postman is used to design, test, and document backend API endpoints before integrating them into the Flutter client. It is particularly useful for verifying authentication flows, inspecting paginated response envelopes, and testing error response shapes against the structured error format expected by the client.

¹³<https://docs.nestjs.com/cli/overview>

¹⁴<https://gitlab.com>



Figure 18: Postman logo

f. Android Emulator / iOS Simulator / Physical Device

Emulators and simulators are used throughout development to validate layout and behavior across Android and iOS screen sizes without requiring physical devices for every test cycle. Physical devices are used specifically for features that require real hardware: camera access for the OCR business card scanning flow, accurate push notification behavior from FCM, and Telegram Mini App rendering within an actual Telegram installation.



Figure 19: Android Studio logo



Figure 20: XCode logo

g. Figma

Figma is used to design and iterate on screen layouts before implementation. UI mockups for the inbox shell, compose view, contact detail screen, and onboarding flow were produced in Figma and used as references during Flutter widget development, reducing back-and-forth between design and implementation.



Figure 21: Figma logo

h. Typst

Typst is used to author this internship report. Its code-based document model allows figures, tables, citations, and cross-references to be managed programmatically, and its fast compilation cycle makes iterative drafting practical throughout the internship period.¹⁵



Figure 22: Typst logo

i. draw.io / Lucidchart / PlantUML

Diagramming tools are used to create architecture, use case, and activity diagrams for the report. These visual materials improve clarity and help explain system structure and workflows more effectively.

draw.io / Lucidchart / PlantUML logo

¹⁵<https://typst.app>

5.5.5. Development Environment Requirements

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VI. IMPLEMENTATION

This chapter documents the concrete implementation of the lorem system. Each subsection corresponds directly to a pattern or architectural decision introduced in [Section III](#) and designed in [Section V](#).

6.1. Project Structure

6.1.1. Backend Monorepo (NestJS)

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```
api/  
├── apps/  
│   ├── api/src/  
│   │   ├── routes/v1/ # stuff  
│   │   └── ... /      # stuff  
│   └── worker/src/  
│       └── ... /      # stuff  
└── libs/  
    └── shared/src/  
        ├── models/    # Sequelize model classes  
        └── domain/     # domain objects
```

Figure 23: Backend monorepo directory structure

Each package has a distinct responsibility, as shown in [Figure 23](#):

- `apps/api` : Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magnam aliquam quaerat voluptatem. Ut enim aequae doleamus animo, cum corpore dolemus, fieri tamen permagna accessio potest, si aliquod aeternum et infinitum impendere malum nobis opinemur. Quod idem licet transferre in voluptatem, ut postea variari voluptas distinguere possit, augeri amplificarique non possit. At.
- `apps/worker` : Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magnam aliquam quaerat voluptatem. Ut

enim aequae doleamus animo, cum corpore dolemus, fieri tamen permagna accessio potest, si aliquod aeternum et infinitum impendere malum nobis opinemur. Quod idem licet transferre.

- `libs/shared` : Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magnam aliquam quaerat voluptatem. Ut enim aequae doleamus animo, cum corpore dolemus, fieri tamen permagna accessio potest, si aliquod aeternum et infinitum impendere.

Table 4: API resource modules and their responsibilities

Module	Responsibility
<code>auth/</code>	Login, logout, OTP, OAuth session management; token and refresh token issuance
<code>auth/register/</code>	New user self-registration flow
<code>auth/session/</code>	Session listing and individual session revocation
<code>auth/telegram/</code>	Telegram Mini App authentication handshake
<code>data/</code>	Read-only reference data: departments, roles, genders, countries
<code>entities/persons/</code>	Person entity CRUD including profile picture and contact handles
<code>entities/companies/</code>	Company entity CRUD including logo upload
<code>mail-accounts/</code>	Link, unlink, and list mail accounts; manage per-account grants
<code>threads/</code>	Thread listing with cursor pagination, unread count, and search
<code>messages/</code>	Fetch, compose, reply to, and forward individual messages
<code>notification-tokens/</code>	Register and deregister FCM device tokens

6.1.2. Mobile Application (Flutter)

The Flutter project uses a layer-first structure under `lib/`, separating infrastructure from feature screens and reusable UI components into four top-level directories: `core/`, `ui/`, `shared/`, and `telegram/`.

```
mobile/lib/
├── core/
│   ├── api/           # HTTP client + 5 interceptors
│   ├── providers/     # global state: auth, theme, locale, notifications
│   ├── repositories/  # data access layer
│   ├── routes/        # GoRouter config, route constants, transitions
│   ├── services/      # deep link handling, haptic feedback
│   ├── storage/       # secure storage, shared preferences, Hive cache
│   ├── models/        # Freezed + json_serializable API response models
│   └── l10n/          # ARB localization files
├── ui/
│   ├── home/          # app scaffold, inbox screen, investor listing
│   ├── login/
│   ├── message/
│   ├── registration/
│   └── splash/
├── shared/
│   ├── components/    # reusable widgets
│   ├── theme/         # Material theme definitions
│   └── utils/
└── telegram/          # Telegram Mini App integration (stub/real split)
```

Figure 24: Flutter project structure under `lib`

Each top-level directory serves a distinct concern, as shown in Figure 24:

- `core/`: All infrastructure independent of any specific screen.
 - **HTTP client**: Stacks five Dio interceptors for token attachment, silent refresh on 401, locale headers, offline detection, and request logging.
 - **Storage**: Separated by security level; `secure_storage.dart` for tokens, `local_storage.dart` for preferences, and `hive_storage.dart` for structured API response cache.
 - **State**: Global `ChangeNotifier` providers registered at the application root; feature-level state lives in per-screen controller classes under `ui/`.
- `ui/`: Feature screens organized by domain. Each subdirectory contains a screen widget, a controller class, and a `widgets/` folder for screen-specific components.
- `shared/`: Reusable widgets, Material theme definitions, and utility functions shared across multiple screens.

- `telegram/`: Telegram Mini App integration behind an abstract interface. `telegram_real.dart` is compiled for the web target and calls the JavaScript WebApp API; `telegram_stub.dart` provides no-op implementations for iOS and Android, so the rest of the application has no platform-conditional branches.

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VII. CONCLUSION

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7.1. Completed Functions

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7.2. Incomplete Functions

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7.3. Challenges

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7.4. Experience

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7.5. Perspective

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7.6. Future Work

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REFERENCES

APPENDICES

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